# Ordinals and Typed Lambda Calculus

## Course Introduction

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# Administrative Information

## Course web page

https://asr.github.io/courses/ordinals-and-typed-lambda-calculus/

## Lectures dates, source code, etc.

See course web page.

# Informally Building Sets

## Definition

A set is **pure** iff its members are also sets.

## Notation

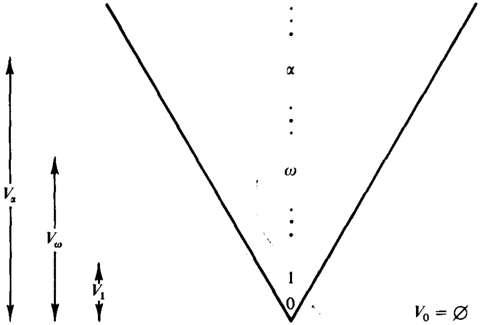
Let $A$ be a set. The power set of $A$ is denoted by $\mathrm{P}A$.

## Convention

We shall use the terms ‘ordinal number’ and ‘ordinal’ interchangeably.

# Informally Building Sets

The ordinal numbers are the backbone of the universe of (pure) sets.[†]




$$V_0 \coloneqq \emptyset \qquad V_\omega \coloneqq V_0 \cup V_1 \cup \cdots \qquad V_{\alpha+1} \coloneqq \mathrm{P}V_\alpha$$

$$V_{n+1} \coloneqq \mathrm{P}V_n \qquad V_{\omega+1} \coloneqq \mathrm{P}V_\omega$$

[†] Figure from (Enderton 1977, Fig. 3).

# Classifying the Ordinals

$$\text{ordinals}\begin{cases}\text{zero ordinal}\\ \text{successor ordinal}\\ \text{limit ordinal}\end{cases}$$

$$\text{ordinals}\begin{cases}\text{countable}\begin{cases}\text{computable ($\lambda$-definable or constructive)}\\ \text{incomputable}\end{cases}\\ \text{uncountable}\end{cases}$$

# References

Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 4).